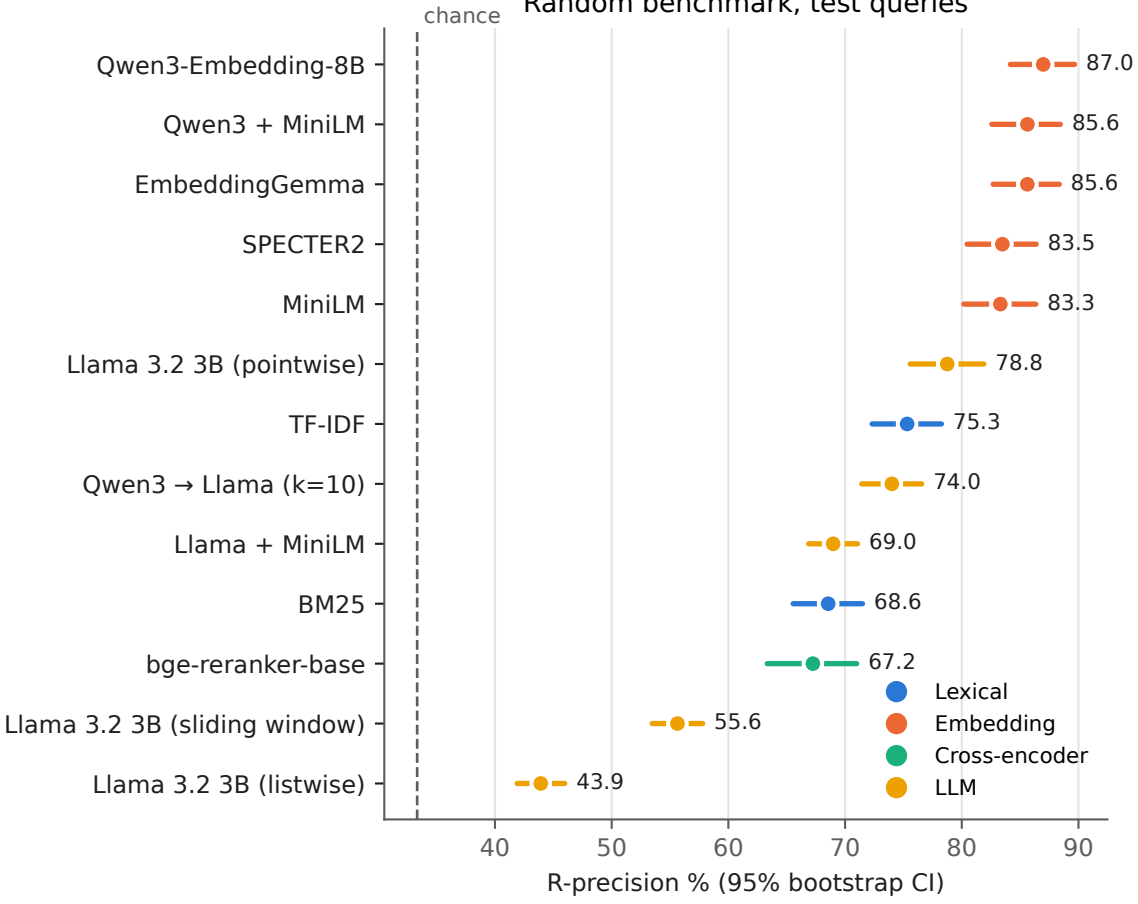
Random benchmark, test queries
chance
Qwen3-Embedding-8B
Qwen3 + MiniLM
EmbeddingGemma
SPECTER2
MiniLM
Llama 3.2 3B (pointwise)
TF-IDF
Qwen3 → Llama (k=10)
Llama + MiniLM
BM25
bge-reranker-base
Llama 3.2 3B (sliding window)
Llama 3.2 3B (listwise)
87.0
85.6
85.6
83.5
83.3
78.8
75.3
74.0
69.0
68.6
67.2
55.6
43.9
Lexical
Embedding
Cross-encoder
LLM
40
50
60
70
80
90
R-precision % (95% bootstrap CI)